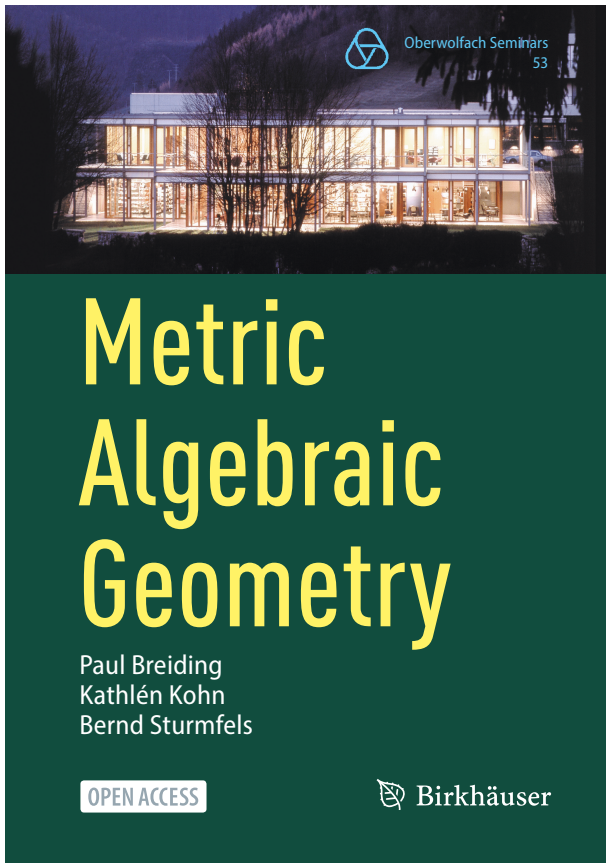
Distance Optimization Within the Grassmannian

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Metric Algebraic Geometry



"…the interplay of metric concepts with algebraic objects."

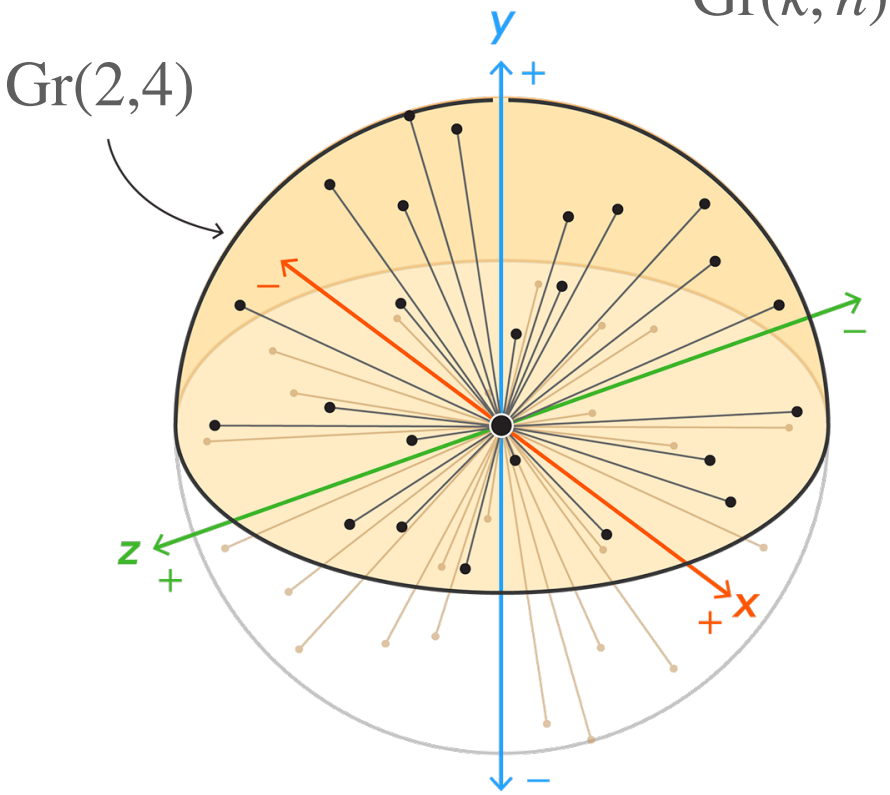
Example.

How does one accurately compute the reach of an algebraic variety?

Today:

Euclidean distance optimization! (Chapter 2)

The Grassmannian



Picture credit: Samuel Velasco/Quanta Magazine

$$\mathrm{Gr}(k,n) = \{k\text{-dimensional subspaces of } \mathbb{R}^n\}$$

$$\sim \{(k-1)\text{-dimensional linear spaces in } \mathbb{P}^{n-1}\}$$

Finding low-dimensional structure of data → Grassmannian optimization

We can represent a vector space with

- a basis or
- an orthogonal projection.

The Grassmannian

L : k-dimensional subspace of $\mathbb{R}^n$ $\quad$ A : $k \times n$ matrix whose rows span L

The $n \times n$ matrix $P = A^T(AA^T)^{-1}A$ is the unique orthogonal projection matrix onto L.

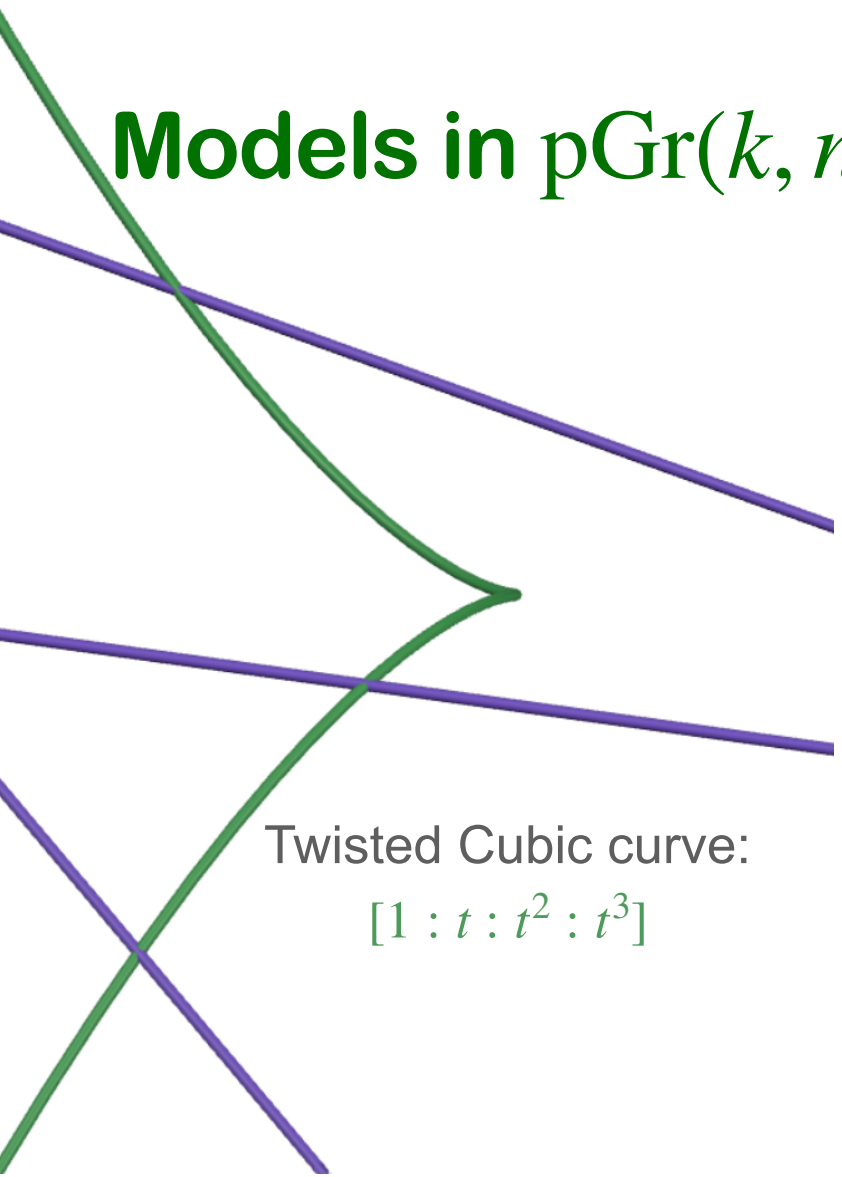
$$\mathrm{pGr}(k,n) = \{P \in \mathrm{Sym}^2\mathbb{C}^n : P^2 = P, \mathrm{trace}(P) = k\}$$

Theorem (Devriendt-F-Reinke-Sturmfels, 2024). The *projection Grassmannian* $\mathrm{pGr}(k,n)$ is a smooth, affine variety of dimension $k(n-k)$. It has prime ideal

$$I(\mathrm{pGr}(k,n)) = \langle P^2 - P, \mathrm{trace}(P) - k \rangle.$$

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{12} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{1n} & p_{2n} & \cdots & p_{nn} \end{pmatrix}$$

Models in $\mathrm{pGr}(k, n)$



Twisted Cubic curve:

$[1 : t : t^2 : t^3]$

A *model* is an irreducible subvariety of $\mathrm{pGr}(k, n)$ defined over $\mathbb{R}$ with Zariski dense real points.

Chow Variety:

$\{$lines in $\mathbb{P}^3$ that meet the twisted cubic curve$\}$

Basis:

$$A = \begin{pmatrix} 1 & t & t^2 & t^3 \\ 0 & 1 & u & v \end{pmatrix}$$

Projection Coordinates:

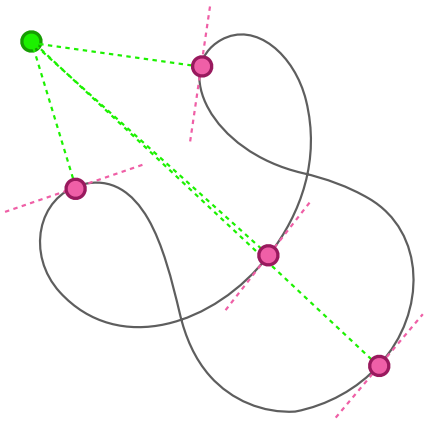
$$\begin{aligned}
&2p_{23}^2p_{24}p_{33} - 2p_{22}p_{24}p_{33}^2 - 2p_{23}^3p_{34} + 2p_{14}p_{22}p_{24}p_{34} - 2p_{13}p_{23}p_{24}p_{34} - 2p_{12}p_{24}^2p_{34} + 2p_{22}p_{23}p_{33}p_{34} \\
&+2p_{12}p_{24}p_{33}p_{34} - 4p_{13}p_{22}p_{34}^2 + 4p_{12}p_{23}p_{34}^2 - 2p_{23}^2p_{34}^2 + p_{11}p_{24}p_{34}^2 + p_{22}p_{24}p_{34}^2 - p_{11}p_{33}p_{34}^2 \\
&+p_{22}p_{33}p_{34}^2 - p_{24}p_{33}p_{34}^2 - p_{33}^2p_{34}^2 + 2p_{23}p_{34}^3 - 2p_{34}^4 - 2p_{14}p_{22}p_{23}p_{44} + 2p_{13}p_{23}^2p_{44} + 2p_{12}p_{23}p_{24}p_{44} \\
&+4p_{13}p_{22}p_{33}p_{44} - 6p_{12}p_{23}p_{33}p_{44} + 2p_{23}^2p_{33}p_{44} - p_{11}p_{24}p_{33}p_{44} - p_{22}p_{24}p_{33}p_{44} + p_{11}p_{33}^2p_{44} \\
&-p_{22}p_{33}^2p_{44} + p_{24}p_{33}^2p_{44} + p_{33}^3p_{44} - 2p_{23}p_{33}p_{34}p_{44} + p_{24}p_{34}^2p_{44} + 3p_{33}p_{34}^2p_{44} - p_{24}p_{33}p_{44}^2 - p_{33}^2p_{44}^2
\end{aligned}$$

Euclidean Distance Optimization

Given $f_1, \ldots, f_r \in \mathbb{R}[P]$, define a model $\mathscr{M} = V(f_1, \ldots, f_r) \cap \mathrm{pGr}(k,n) \subseteq \mathrm{Sym}^2\mathbb{C}^n$.

Given a data point $Q \in \mathrm{Sym}^2\mathbb{R}^n$, what point on $\mathscr{M}_{\mathbb{R}}$ is closest to Q?

$$\min_{P \in \mathscr{M}_{\mathbb{R}}} ||P - Q||_F^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} (p_{ij} - q_{ij})^2 \qquad \mathscr{J}^Q = \begin{pmatrix} \nabla ||P - Q||_F^2 \\ \mathrm{Jac}_{\mathrm{pGr}(k,n)}(P) \\ -\nabla f_1(x)- \\ \vdots \\ -\nabla f_r(x)- \end{pmatrix}$$



Critical points:

$$\{P \in \mathscr{M}_{\mathrm{reg}} : \mathrm{rank}(\mathscr{J}^Q) = \mathrm{codim}(\mathscr{M})\}$$

The Euclidean Distance Degree

Definition/Proposition (Draisma-Horobeț-Ottaviani-Sturmfels-Thomas, 2016).
The *Euclidean distance (ED) degree* of a model $\mathscr{M}$ is the cardinality of the set

$$\overline{\{P \in \mathscr{M}_{\text{reg}} : \operatorname{rank}(\mathscr{J}^Q) = \operatorname{codim}(\mathscr{M})\}}$$

for generic $Q \in \operatorname{Sym}^2\mathbb{R}^n$. The ED degree is finite.

Proof Idea.

$$\overline{\{(P,Q) \in \mathscr{M}_{\text{reg}} \times \operatorname{Sym}^2\mathbb{C}^n : \operatorname{rank}(\mathscr{J}^Q) = \operatorname{codim}(\mathscr{M})\}}$$



The Euclidean Distance Degree of $\mathrm{pGr}(k,n)$

Theorem (Lai-Lim-Ye, 2025). The ED degree of $\mathrm{pGr}(k,n)$ is $\binom{n}{k}$.

Proposition (FHR). For any $Q \in \mathrm{Sym}^2\mathbb{C}^n$, the critical point set of

$$\min_{P \in \mathrm{pGr}(k,n)} ||P-Q||_F^2$$

$$\mathscr{J}^Q = \begin{pmatrix} \nabla ||P-Q||_F^2 \\ \mathrm{Jac}_{\mathrm{pGr}(k,n)}(P) \end{pmatrix}$$

is $\{P \in \mathrm{pGr}(k,n) : PQ = QP\}$.

Theorem (FHR). The data point $Q \in \mathrm{Sym}^2\mathbb{C}^n$ has $\binom{n}{k}$ critical points if and only if Q has n distinct eigenvalues.

In particular, a data point $Q \in \mathrm{pGr}(k,n)$ has infinitely many critical points.

Extraneous Critical Points

Given $f_1, \ldots, f_r \in \mathbb{R}[P]$, define a model $\mathscr{M} = V(f_1, \ldots, f_r) \cap \mathrm{pGr}(k,n) \subseteq \mathrm{Sym}^2\mathbb{C}^n$.

Suppose now that $Q \in \mathrm{pGr}(k,n)$ (instead of $\mathrm{Sym}^2\mathbb{R}^n$).

The incidence correspondence

$$\overline{\{(P,Q) \in \mathscr{M}_{\mathrm{reg}} \times \mathrm{pGr}(k,n) : \mathrm{rank}(\mathscr{J}^Q) = \mathrm{codim}(\mathscr{M})\}}$$

is no longer irreducible!

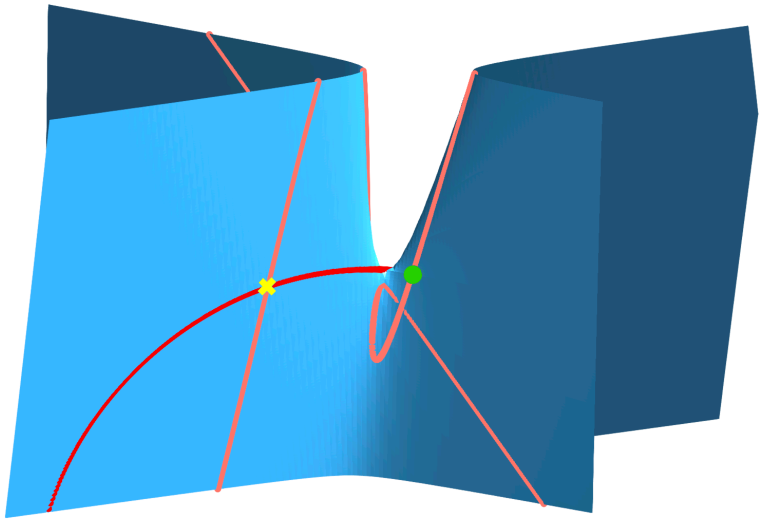
$$\mathscr{J}^Q = \begin{pmatrix} \nabla \,||\, P - Q \,||_F^2 \\ \mathrm{Jac}_{\mathrm{pGr}(k,n)}(P) \\ -\nabla f_1(x)- \\ \vdots \\ -\nabla f_r(x)- \end{pmatrix}$$

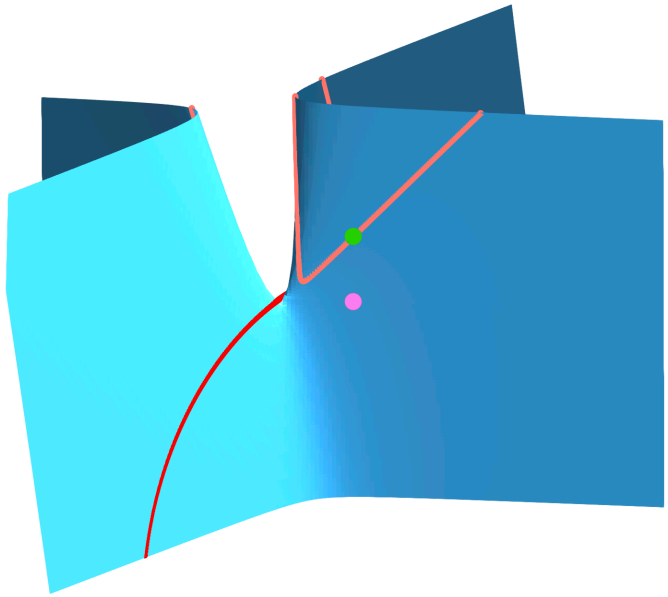
Proposition (FHR). The matrix $\mathscr{J}^Q$ drops rank at every point in $\mathscr{E}^Q_{\mathscr{M}} = \overline{\{P \in \mathscr{M}_{\mathrm{reg}} : PQ = QP\}}$.

Extraneous Critical Points

The algebraic cut locus of Q is $\operatorname{cut}(Q) = \{P \in \mathscr{M} : \operatorname{rank}(PQ) < k\}$.

Proposition (FHR). $\mathscr{E}^Q_{\mathscr{M}} = \overline{\{P \in \mathscr{M}_{\text{reg}} : PQ = QP\}} \subseteq \operatorname{cut}(Q)$.





The Grassmann Distance Degree

Definition/Proposition (FHR). The *Grassmann distance (GD) degree* of a model $\mathscr{M}$ is the cardinality of the set

$$\overline{\{P \in \mathscr{M}_{\text{reg}} \backslash \text{cut}(Q) : \text{rank}(\mathscr{J}^Q) = \text{codim}(\mathscr{M})\}}$$

for generic $Q \in \text{pGr}(k,n)$.

Theorem (FHS). Every model in $\text{pGr}(2,n)$ has finite GD degree.

Remark. The incidence correspondence

$$\overline{\{(P,Q) \in \mathscr{M}_{\text{reg}} \backslash \text{cut}(Q) \times \text{pGr}(k,n) : \text{rank}(\mathscr{J}^Q) = \text{codim}(\mathscr{M})\}}$$

is still not necessarily irreducible, e.g., the curve parametrized by $A = \begin{pmatrix} 1 & 0 & t & 0 \\ 0 & 1 & 0 & t \end{pmatrix}$.

Matroid Varieties

Matroid Varieties

A (realizable) rank k matroid on n elements encodes dependencies among columns of a $k \times n$ matrix. For $k = 2$, the realization space is

$$V_{\pi,\ell} = \left\{ \begin{bmatrix} A_1 & A_2 & \cdots & A_r & 0 \end{bmatrix} \in \mathbb{C}^{2\times n} : A_i \in \mathbb{C}^{2\times \pi_i},\ \operatorname{rank}(A_i) = 1 \right\}.$$

For $0 \le \ell \le n$ and $\pi \vdash n - \ell$, let $\mathscr{V}_{\pi,\ell}$ be the set of projection matrices onto vector spaces with bases in $V_{\pi,\ell}$.

Theorem (FHR). The GD degree of $\mathscr{V}_{\pi,\ell} \subseteq \mathrm{pGr}(2,n)$ is

$$\begin{cases} 2 & \text{if } r = 2 \text{ and } \ell = 0, \text{ and} \\ 2^{\#\{i:\pi_i \ge 2\}} & \text{otherwise.} \end{cases}$$

Matroid Varieties

We say $\pi \leq \pi'$ if $\pi_i \leq \pi'_i$ for all $i = 1, \ldots, r$.

Theorem (FHR). If $\pi \leq \pi'$, then the ED degree of $\mathscr{V}_{\pi,\ell}$ is at most the ED degree of $\mathscr{V}_{\pi',\ell}$.

Partition π	1^4	21^2	2^2	31	1^5	21^3	2^21	31^2	32	41
ED Degree of $\mathscr{V}_\pi$	6	10	4	3	10	24	44	16	6	4

Partition π	1^6	21^5	2^21^2	2^3	31^3	321	3^2	41^2	42	51
ED Degree of $\mathscr{V}_\pi$	15	44	120	168	39	76	9	22	8	5

Schubert Varieties

Schubert Varieties

$$\mathcal{S}_{i_1 \cdots i_k} = \{A : \text{the reduced row echelon form of } A \text{ has pivots } i_1, \ldots, i_k\}$$

$$\begin{pmatrix} 1 & 0 & 0 & a & b & c \\ 0 & 1 & 0 & d & e & f \\ 0 & 0 & 1 & g & h & i \end{pmatrix} \qquad \begin{pmatrix} 1 & a & 0 & b & 0 & c \\ 0 & 0 & 1 & d & 0 & e \\ 0 & 0 & 0 & 0 & 1 & f \end{pmatrix} \qquad \begin{pmatrix} 1 & a & b & c & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Theorem (FHR). The GD degree of $\mathcal{S}_{i_1 \cdots i_k}$ is 1 if $i_{j+1} = (i_j \mod n) + 1$ for $j = 1, \ldots, n-1$.

Schubert Varieties

When $k = 2$, the Schubert variety $\mathcal{S}_{ij}$ is realized by matrices of the form

$$\begin{pmatrix} 0 & \cdots & 0 & 1 & * & \cdots & * & 0 & * & \cdots & * \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 1 & * & \cdots & * \end{pmatrix}$$
$$\qquad\qquad\qquad\quad i \qquad\qquad\qquad\quad j$$

Theorem (Lai-Lim-Ye, 2025). The ED degrees of certain Schubert varieties are

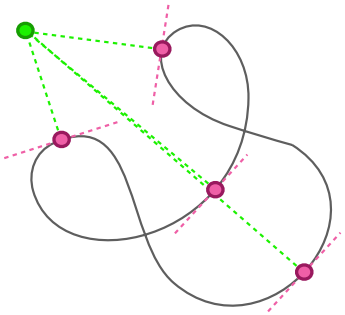
$$\mathrm{EDdegree}(\mathcal{S}_{i,i+1}) = \binom{n-i+1}{2} \qquad\qquad \mathrm{EDdegree}(\mathcal{S}_{in}) = n - i\,.$$

Conjecture (F-Rosana-Sturmfels, 2026). If $i < j - 1$, then

$$\mathrm{EDdegree}(\mathcal{S}_{ij}) = 2(n-j+1)^2(j-i) - (n-j+1)(n-i)$$

Theorem (FHR). The Schubert varieties $\mathcal{S}_{1n}$ and $\mathcal{S}_{i,i+1}$ GD degree 1. All other Schubert varieties in $\mathrm{pGr}(2,n)$ have GD degree 2.

Thank you!



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